

FYP Prep Plan

Mon 20 July - Sun 9 August 2026 | three weeks before the FYP formally starts

Budget: 1-2 h per weekday (working), 4 h per weekend day - roughly 45-50 usable hours.

Goal of these three weeks: de-risk the real project. Build Architecture A end to end in MuJoCo - the detector, depth maths, transforms, primitives and LLM planner are the *real* FYP code; only the image source and controller backend change when hardware arrives. Clear the reading backlog. Hand over a consolidated repo and a refined proposal so day one of school is implementation, not setup. The FYP itself remains a real-robot comparison; simulation here is the development substrate, not a deliverable.

Ground rules: papers are commute reading; weekday code slots are short, so keep each to one clear task; weekend blocks absorb overflow from the week.

WEEK 1 . 20-26 Jul 2026 - Unblock, read SmolVLA, build the scene, start perception

Mon 20 Jul (1-2 h) - Email Prof Cheah (do this first)

- ☐ Ask: can development and iteration run in simulation, with the physical UR5e used for the final A-vs-B evaluation?
- ☐ Ask: is a semester-1 (end-November) completion acceptable?
- ☐ Ask: what are the fixed assessment milestones (interim report, final report, presentation dates)?
- ☐ Ask: when is lab / UR5e / RGB-D camera access available, and is there a budget for parts and cloud compute?

Longest-latency item, and its answer changes what you build. Do not let this slip past day one.

Tue 21 Jul (1-2 h) - Fix the DemoRecorder timestamp

- ☐ Pass MuJoCo's `data.time` into `DemoRecorder.record()` instead of `time.monotonic()`
- ☐ Re-record a short episode
- ☐ Re-run the converter and confirm it reports `fps = 20`

config.yaml already has `record_hz: 20` and `RECORD_EVERY = 25`, which is correct in simulation time. The bug is wall-clock stamping while the sim runs ~4x slower than real time, so 20 Hz is logged as ~5 Hz.

Wed 22 Jul (1-2 h) - SmolVLA paper, first half

- ☐ Sections 1-3: ~450M params vs 7B OpenVLA
- ☐ SmolVLM-2 backbone plus the action expert
- ☐ Async inference

Commute reading.

Thu 23 Jul (1-2 h) - SmolVLA paper, second half

- ☐ Sections 4-6: training data and key ablations
- ☐ Write the 1-page summary while it is fresh

Fri 24 Jul (1-2 h) - MuJoCo scene: add objects

- ☐ Add 3-4 graspable objects (coloured blocks) to `scene_gripper.xml`
- ☐ Give each a free joint so it is dynamic
- ☐ Set sensible mass, friction and collision geoms

Start simple. Contact tuning is a genuine time sink.

Sat 25 Jul (4 h) - Finish the scene

- ☐ Verify the gripper can close on and lift each object
- ☐ Add a bin or target region
- ☐ Script randomised object placement

Shared infrastructure: Architecture A detects these objects, Architecture B demonstrates on them, and the Tier A/B/C evaluation needs the variety.

Sun 26 Jul (4 h) - Open-vocabulary detection

- ☐ Run OWL-ViT or GroundingDINO on a rendered frame
- ☐ Text query to boxes + confidence; set a threshold
- ☐ Visualise the overlays

Watch model download size and VRAM usage alongside MuJoCo.

WEEK DELIVERABLES

- ☐ Email sent to Prof Cheah
- ☐ Recorder timestamps correct at 20 Hz
- ☐ SmolVLA read + 1-page summary written
- ☐ MuJoCo scene with graspable objects
- ☐ Detector returning boxes on rendered frames

WEEK 2 . 27 Jul - 2 Aug 2026 - Geometry, primitives, LLM

Mon 27 Jul (1-2 h) - Depth to 3D

- ☐ Finish the MuJoCo depth render
- ☐ Convert box-centre pixel + depth + camera intrinsics to camera-frame XYZ
- ☐ Test against a known object position

Tue 28 Jul (1-2 h) - Hand-eye calibration (free win in sim)

- ☐ Build the camera-to-base transform with pose_to_T / pose_trans
- ☐ Transform a detected object into the base frame
- ☐ Verify against MuJoCo's ground-truth body position

You know the fixed camera pose exactly in sim, so the whole geometry chain can be validated before facing real calibration.

Wed 29 Jul (1-2 h) - Open X-Embodiment paper

- ☐ Padalkar et al. 2023: action and observation formats
- ☐ Check UR5 inclusion

The last outstanding Week-4 reading item.

Thu 30 Jul (1-2 h) - euler_to_rotvec + the pick primitive

- ☐ Add euler_to_rotvec (and euler_to_R) to transforms.py with round-trip tests
- ☐ pick(xyz): approach above, descend, close, lift

euler_to_rotvec is needed to turn planned or predicted Euler orientations back into controller commands.

Fri 31 Jul (1-2 h) - The place primitive

- ☐ place(xyz): above, descend, open, retract
- ☐ Test a full hardcoded pick-and-place in MuJoCo

Build both on the shared controller interface so nothing needs rewriting for hardware.

Sat 1 Aug (4 h) - Anthropic SDK + tool calling

- ☐ API key and a hello-world call
- ☐ Read the tool-use docs
- ☐ Define pick / place / move as tool schemas
- ☐ Note rate limits and cost per call

Sun 2 Aug (4 h) - LLM planner over primitives

- ☐ System prompt listing available skills plus detected objects
- ☐ Model returns an ordered tool-call plan
- ☐ Parse the plan and dispatch to your primitives
- ☐ Test 'put the red block in the bin' with real detections feeding the prompt

WEEK DELIVERABLES

- ☐ Depth-to-3D and hand-eye validated against ground truth
- ☐ pick / place primitives working in MuJoCo
- ☐ OXE paper read
- ☐ LLM planner producing executable tool-call plans

WEEK 3 . 3-9 Aug 2026 - Closed loop, consolidation, handover

Mon 3 Aug (1-2 h) - Wire the full slice (open loop)

- ☐ frame -> detect -> depth-to-3D -> base frame -> LLM plan -> execute in MuJoCo

Open-loop first; the re-detect step comes later in the week.

Tue 4 Aug (1-2 h) - Debug end to end

- ☐ Get one full task running reliably
- ☐ Log failures by stage: detection / depth / transform / planning / execution

Per-stage diagnosability is the entire argument for Architecture A, so make the logging good. It becomes evidence in your write-up.

Wed 5 Aug (1-2 h) - RTC paper

- ☐ Black et al. 2025: async inference and action chunking

Connects directly to the `chunk_size = 50` seen in the SmolVLA config.

Thu 6 Aug (1-2 h) - Close the loop

- ☐ Add re-detection before each grasp
- ☐ Document the failure modes you hit

Fri 7 Aug (1-2 h) - Refine the proposal

- ☐ Re-read end to end; check NTU EEE formatting
- ☐ Update the architecture diagram
- ☐ Tighten the methodology to reflect what you now know

Sat 8 Aug (4 h) - Consolidate the codebase

- ☐ Update the README
- ☐ A runnable demo each for URController, DemoRecorder, detector, planner
- ☐ Tidy the repo; make sure the tests pass

Sun 9 Aug (4 h) - Buffer and handover

- ☐ Absorb slippage from the previous three weeks
- ☐ Write a 1-page current-state summary
- ☐ Write a priority list for FYP weeks 1-2
- ☐ Agree a review cadence with your supervisor
- ☐ STRETCH (only if genuinely ahead): Architecture B inference loop driving the MuJoCo arm

WEEK DELIVERABLES

- ☐ Architecture A closed-loop slice running in simulation
 - ☐ RTC paper read
 - ☐ Proposal refined and ready to send
 - ☐ Repo consolidated: all four components runnable
 - ☐ 1-page summary + FYP week-1 priority list
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IF YOU FALL BEHIND - drop from the bottom up

1. Email the professor - never skip, never delay past day one
2. Architecture A slice (detection, depth, transform, primitives, planner) - the core de-risking
3. Recorder timestamp fix - small, but blocks any data you intend to keep
4. Papers - SmolVLA first, then RTC, then OXE
5. Proposal refinement and consolidation
6. Architecture B inference loop - stretch goal, drop freely

DELIBERATELY NOT IN THIS PLAN

- A 50-episode sim collection - those demos will not appear in your thesis; the real comparison uses real UR5e demonstrations
- Any long fine-tune - the pipeline is already proven with a 1-step dry run; a real run is semester work on cloud GPU
- Sim evaluation metrics or success rates - the deliverable is 'both pipelines execute', not measured results
- Sim-to-real transfer work - out of scope for three weeks

ALREADY DONE (as of 19 Jul 2026)

- ☒ src/fyp/transforms.py - full frame-math layer (quat_to_rotvec, rotvec_to_R, R_to_rotvec, rotvec_to_euler, pose_to_T, T_to_pose, pose_trans, pose_inv), verified to machine precision
- ☒ scripts/hdf5_to_lerobot.py - HDF5 to LeRobot converter, validated on real episodes (227 frames, 2 episodes)
- ☒ SmolVLA training pipeline proven - 1-step fine-tune ran against the converted dataset on the local GPU